

Plant Disease Classification Using Transfer Learning

1. Main Objective of the Analysis

The objective of this project is to develop an accurate deep learning model that automatically classifies plant leaf diseases from images. The analysis focuses on transfer learning using Convolutional Neural Networks (CNNs). Three pretrained CNN architectures—ResNet50, DenseNet121, and EfficientNetB0—are compared to determine which provides the best classification performance. The goal is to support farmers and agricultural stakeholders through faster and more reliable disease diagnosis, enabling timely treatment and reducing crop losses.

2. Deep Learning Technique Used

This project uses supervised Deep Learning. Specifically, transfer learning with pretrained Convolutional Neural Networks (CNNs) is employed. ResNet50 is trained first using ImageNet pretrained weights. Training is performed in two stages: (1) training a custom classifier while freezing the convolutional backbone, and (2) fine-tuning the last 50 layers using a lower learning rate to adapt the pretrained features to plant disease images.

3. Dataset Description

The PlantVillage dataset contains 54,305 images belonging to 38 plant disease categories across multiple crop species. For this project, 43,444 images were used for training and 10,861 images for validation. Images were resized to 224×224 pixels and categorical labels were generated using TensorFlow ImageDataGenerator.

4. Data Exploration and Preprocessing

Dataset exploration showed that several classes were imbalanced. Class weights were therefore calculated using scikit-learn and supplied during training. Data augmentation included random rotations, zooming, horizontal flipping, and width/height shifts. Images were preprocessed using the ResNet50 `preprocess_input` function.

5. Model Variations

Variation 1: ResNet50 with frozen convolutional layers.

Variation 2: ResNet50 fine-tuned by unfreezing the last 50 layers.

Variation 3: DenseNet121 and EfficientNetB0 are planned for comparison using the same preprocessing and evaluation pipeline. The final report will compare all three architectures.

6. Results

Stage 1 (Frozen ResNet50): Training Accuracy = 93.27%, Validation Accuracy = 95.84%, Validation Loss = 0.1216.

Stage 2 (Fine-Tuned ResNet50): Training Accuracy = 97.65%, Validation Accuracy = 98.74%, Validation Loss = 0.0370.

Fine-tuning improved validation accuracy by approximately 2.9 percentage points.

7. Recommended Model

Based on the current experiments, the fine-tuned ResNet50 model is recommended because it achieved 98.74% validation accuracy while maintaining a low validation loss. The final recommendation will be updated after DenseNet121 and EfficientNetB0 experiments are completed.

8. Key Findings

Transfer learning significantly reduced training time and achieved high accuracy. Fine-tuning substantially improved performance over the frozen model. Handling class imbalance with class weights contributed to robust performance across disease categories. The confusion matrix showed strong diagonal dominance, indicating accurate predictions for most classes.

9. Challenges and Solutions

Challenge: Class imbalance. Solution: Computed and applied class weights.

Challenge: Risk of overfitting. Solution: Used dropout, data augmentation, EarlyStopping, and ReduceLROnPlateau.

Challenge: Adapting generic ImageNet features. Solution: Fine-tuned the last 50 layers with a learning rate of $1e-5$.

10. Future Work

Complete DenseNet121 and EfficientNetB0 experiments, compare all three models using Accuracy, Precision, Recall, and F1-score, incorporate Grad-CAM for explainability, evaluate on real-world field images, and deploy the best model as a web or mobile application.